

Prime Factors, HCF and LCM

Break every number into primes. HCF = what they share (lowest powers). LCM = everything (highest powers).



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THE TOPIC ON ONE MAP

Prime Factors, HCF and LCM

01 Product of prime factors

1400 as powers of primes

3.6×10^3

smallest multiplier for a square

02 HCF and LCM of numbers

HCF of 56 and 84

LCM of 28, 42, 63

LCM of $5A$ and $7B$

03 HCF and LCM in index form

$A = 2^8 \times 3^5 \times 11^4$

unknown powers m, n, p

$P^3 \times Q$

04 LCM problems: “together again”

buses, trams, lamps

packs of pens and pencils

next date both

WHICH ONE? Read the story first:

ZONE HCF

Biggest that divides both

shares out / splits into equal groups:
lowest powers of the **common** primes

ZONE LCM

Smallest both divide into

happens together again / same number
of each: highest power of **every** prime

01 Product of Prime Factors

Split the number into any two factors, keep splitting until every branch ends in a prime. Then write the primes in order, as powers.

WHEN YOU SEE → YOU DO

“as a product of its prime factors”

→ factor tree (or ladder), then multiply the primes

“as a product of powers of its prime factors”

→ index form: $2^3 \times 5^2 \times 7$

“Show your working clearly”

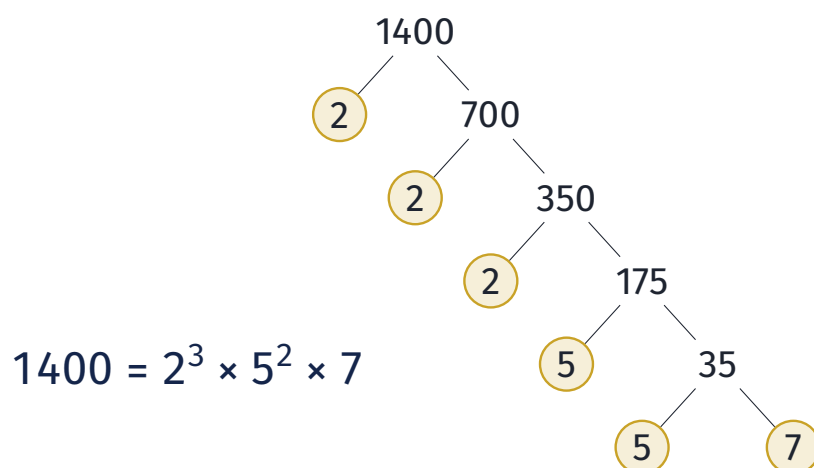
→ the tree / ladder is the method mark

3.6×10^3 , 480×10^9

→ split: 36×100 — then $10 = 2 \times 5$

“smallest number to multiply by to make a square”

→ make every power even



× **TRAP** Stopping at 35 or 25: every end of the tree must be a **prime** (circled).

IDEA 01

Practise: Product of Prime Factors

My questions first, then selected practice, then real past papers.

Q1 DR ESLAM Factor tree

CH 1 • TOPIC 3 2 marks

Write 90 as a product of powers of its prime factors.

Q2 DR ESLAM Index form

CH 1 • TOPIC 3 2 marks

Write 360 as a product of powers of its prime factors.

SELECTED PRACTICE

Q3

PRACTICE

Practice • Easy

CH 1 • TOPIC 3

3 marks

- 1 Write 525 as a product of its prime factors.

(3 marks)

Q4

PRACTICE

Practice • Easy

CH 1 • TOPIC 3

2 marks

6 $A = 2^n \times 3 \times 5^m$

Write $8A$ as a product of powers of its prime factors.

(2 marks)

Q5**PRACTICE**

Practice • Easy

CH 1 • TOPIC 3 3 marks

14 Write 504 as the product of its prime factors.

(3 marks)

Q6**PRACTICE**

Practice • Hard

CH 1 • TOPIC 3 3 marks

10 N is a number.

As a product of prime factors in index form $N = 2 \times 3^4 \times y^3$

Work out $3N^2$ as a product of prime factors in index form.

Give your answer in terms of y .

(3 marks)

PAST PAPERS

Q7**PAST PAPER**

Edexcel 4MA1 • January 2020 • Paper 2HR • Q1

CH 1 • TOPIC 3 3 marks

You must write down all the stages in your working.

- 1 (a) Write $5^{17} \times 5^2$ as a single power of 5

.....

- (b) Write 800 as a product of its prime factors.
Show your working clearly.

.....

(Total for Question 1 is 3 marks)

Q8**PAST PAPER**

Edexcel 4MA1 • May 2021 • Paper 2H • Q1

CH 1 • TOPIC 3 3 marks

You must write down all the stages in your working.

- 1 Write 600 as a product of powers of its prime factors.
Show your working clearly.

.....

(Total for Question 1 is 3 marks)

Q9**PAST PAPER**

Edexcel 4MA1 • November 2021 • Paper 1H • Q7

CH 1 • TOPIC 3 3 marks

- 7 Write 3.6×10^3 as a product of powers of its prime factors.
Show your working clearly.
-

(Total for Question 7 is 3 marks)

Q10**PAST PAPER**

Edexcel 4MA1 • June 2022 • Paper 2HR • Q2

CH 1 • TOPIC 3 3 marks

- 2 Write 1200 as a product of powers of its prime factors.
Show your working clearly.
-

(Total for Question 2 is 3 marks)

Q11**PAST PAPER**

Edexcel 4MA1 • May 2023 • Paper 1HR • Q1

CH 1 • TOPIC 3**3 marks**

You must write down all the stages in your working.

- 1 Write 2250 as a product of powers of its prime factors.
Show your working clearly.

.....
(Total for Question 1 is 3 marks)

Q12**PAST PAPER**

Edexcel 4MA1 • June 2024 • Paper 2HR • Q1

CH 1 • TOPIC 3**3 marks**

You must write down all the stages in your working.

- 1 Write 1400 as a product of powers of its prime factors.
Show your working clearly.

.....
(Total for Question 1 is 3 marks)

Q13**PAST PAPER**

Edexcel 4MA1 • June 2025 • Paper 2HR • Q2

CH 1 • TOPIC 3 3 marks

- 2 Write 1400 as a product of powers of its prime factors.
Show your working clearly.

.....

(Total for Question 2 is 3 marks)

02 HCF and LCM of Numbers: the Venn Method

Write both numbers as prime factors. Shared primes go in the middle. HCF = multiply the middle. LCM = multiply **everything** in the diagram.

WHEN YOU SEE → YOU DO

“highest common factor (HCF)”

→ multiply the overlap

“lowest common multiple (LCM)”

→ multiply every number in the Venn diagram

three numbers: “LCM of 28, 42 and 63”

→ highest power of each prime from any of them

“circle the LCM”

→ check it is divisible by every number

Prime factors

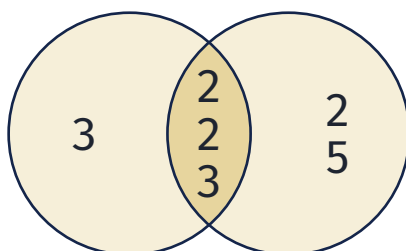
Venn diagram

HCF = middle

LCM = all

$$36 = 2^2 \times 3^2$$

$$120 = 2^3 \times 3 \times 5$$



$$\text{HCF} = 2 \times 2 \times 3 = 12$$

$$\text{LCM} = 3 \times 2 \times 2 \times 3 \times 2 \times 5 = 360$$

CALCULATOR CHECK The calculator has FACT (prime factors) — use it to check your tree, never instead of the working.

IDEA 02

Practise: HCF and LCM of Numbers

My questions first, then selected practice, then real past papers.

Q14 DR ESLAM Venn method

CH 1 • TOPIC 3 3 marks

Find (a) the HCF (b) the LCM of 48 and 180.

SELECTED PRACTICE**Q15** PRACTICE Practice • Easy

CH 1 • TOPIC 3 2 marks

11 Find the Highest Common Factor (HCF) of 140 and 245

(2 marks)

Q16 **PRACTICE** Practice • Medium

CH 1 • TOPIC 3 2 marks

- 5 Find the highest common factor (HCF) of 72 and 90.
(2 marks)

Q17 **PRACTICE** Practice • Medium

CH 1 • TOPIC 3 3 marks

- 14 (a) Written as the product of its prime factors

$$672 = 2^5 \times 3 \times 7$$

Write 252 as the product of its prime factors.

(2 marks)

- (b) Work out the value of the highest common factor of 672 and 252

(1 mark)

Q18**PRACTICE**

Practice • Hard

CH 1 • TOPIC 3 4 marks

13 Here are some properties of a number.

- It is a common factor of 288 and 360.
- It is a common multiple of 4 and 6.
- It is larger than 25.

Find the **two** possible numbers with these properties.

(4 marks)

PAST PAPERS

Q19**PAST PAPER**

Edexcel 4MA1 • January 2020 • Paper 1H • Q2

CH 1 • TOPIC 3 2 marks

2 Find the lowest common multiple (LCM) of 28 and 105

(Total for Question 2 is 2 marks)

Q20**PAST PAPER**

Edexcel 4MA1 • January 2022 • Paper 2H • Q7

CH 1 • TOPIC 3 3 marks

- 7 Find the lowest common multiple (LCM) of 28, 42 and 63
Show your working clearly.

(Total for Question 7 is 3 marks)

Q21**PAST PAPER**

Edexcel 4MA1 • June 2022 • Paper 1H • Q3

CH 1 • TOPIC 3 4 marks

- 3 (a) Find the highest common factor (HCF) of 56 and 84
Show your working clearly.
- (b) Find the lowest common multiple (LCM) of 60 and 72
Show your working clearly.

(Total for Question 3 is 4 marks)

Q22**PAST PAPER**

Edexcel 4MA1 • May 2023 • Paper 1H • Q8

CH 1 • TOPIC 3 4 marks

- 8 (a) Write 300 as a product of its prime factors.
Show your working clearly.

$$A = 2 \times 2 \times 2 \times 3 \times 3 \times 5$$

$$B = 2 \times 2 \times 3 \times 3 \times 3 \times 5$$

- (b) Find the lowest common multiple (LCM) of $5A$ and $7B$
Show your working clearly.

(Total for Question 8 is 4 marks)

Q23**PAST PAPER**

Edexcel 4MA1 • May 2024 • Paper 1H • Q3

CH 1 • TOPIC 3 2 marks

- 3 Find the highest common factor (HCF) of 72 and 108
Show your working clearly.

(Total for Question 3 is 2 marks)

Q24**PAST PAPER**

Edexcel 4MA1 • June 2025 • Paper 1HR • Q1

CH 1 • TOPIC 3 2 marks**You must write down all the stages in your working.**

- 1 Find the lowest common multiple (LCM) of 45 and 70

.....

(Total for Question 1 is 2 marks)

03 HCF and LCM in Index Form

No calculation needed. Line the primes up. HCF: **lowest** power of each **shared** prime. LCM: **highest** power of **every** prime.

WHEN YOU SEE → YOU DO

$$A = 2^8 \times 3^5 \times 11^4, B = 2^6 \times 3 \times 11^8$$

$$\longrightarrow \text{HCF } 2^6 \times 3 \times 11^4 \cdot \text{LCM } 2^8 \times 3^5 \times 11^8$$

a prime in only one number

→ in the LCM, never in the HCF

“HCF of 5A and 2B”

→ first update the powers: 5A adds 1 to the power of 5

unknown powers with $m > 2$

→ use the inequality to see which power is smaller

“find p, q, r ” from a given HCF/LCM

→ match each prime’s power one at a time

	2	3	5	7
$A = 3^2 \times 5^4 \times 7$	—	3^2	5^4	7
$B = 3^4 \times 5^3 \times 7 \times 11$	—	3^4	5^3	7
HCF (lowest)		3^2	5^3	7
LCM (highest, + 11)		3^4	5^4	7×11

× **TRAP** Adding the powers (that is multiplying $A \times B$), not choosing the lowest or highest one.

IDEA 03

Practise: HCF and LCM in Index Form

My questions first, then selected practice, then real past papers.

Q25 DR ESLAM Lowest, highest

CH 1 • TOPIC 3 2 marks

$A = 2^4 \times 3 \times 5^2$, $B = 2^2 \times 3^3 \times 7$. Find the HCF and the LCM of A and B .

SELECTED PRACTICE**Q26** PRACTICE Practice • Medium

CH 1 • TOPIC 3 4 marks

7 (a) $A = 3^2 \times 5^4 \times 7$ $B = 3^4 \times 5^3 \times 7 \times 11$

Find the highest common factor (HCF) of A and B .

(2 marks)

(b) Find the lowest common multiple (LCM) of A and B .

(2 marks)

Q27**PRACTICE**

Practice • Medium

CH 1 • TOPIC 3 7 marks

17 (a) i) Write 120 as a product of its prime factors.

[3]

ii) The lowest common multiple (LCM) of x and 120 is 360. Find the smallest possible value of x .

[2]

(5 marks)

(b) Two numbers, A and B , are written as a product of prime factors.

$$A = 2^4 \times 3^2 \times 7^2 \quad B = 2^3 \times 3 \times 5 \times 7$$

Find the highest common factor (HCF) of A and B .

(2 marks)

Q28**PRACTICE**

Practice • Hard

CH 1 • TOPIC 3 2 marks

11 You are given that

$$90 = 3^2 \times 2 \times 5 \text{ and } 177147 = 3^{11}$$

Find the lowest common multiple (LCM) of 270 and 177147. Give your answer using power notation and as an ordinary number.

using power notation

as an ordinary number

(2 marks)**Q29****PRACTICE**

Practice • Hard

CH 1 • TOPIC 3 1 mark

14 Here is a list of five numbers.

$$98^{53} \quad 98^{64} \quad 98^{73} \quad 98^{88} \quad 98^{91}$$

Find the lowest common multiple of these five numbers.

(1 mark)

PAST PAPERS

Q30**PAST PAPER**

Edexcel 4MA1 • Oct/Nov 2020 • Paper 1H • Q10

CH 1 • TOPIC 3 3 marks

$$10 \quad A = 2 \times 3^{43}$$

$$B = 16 \times 3^{37}$$

(a) Find the highest common factor (HCF) of A and B .

.....

(b) Express the number $A \times B$ as a product of powers of its prime factors.
Give your answer in its simplest form.

.....

(Total for Question 10 is 3 marks)

Q31**PAST PAPER**

Edexcel 4MA1 • January 2021 • Paper 1HR • Q9

CH 1 • TOPIC 3 4 marks

$$9 \quad A = 2^8 \times 3^5 \times 11^4 \quad B = 2^6 \times 3 \times 11^8$$

(a) Find the highest common factor (HCF) of A and B .

.....

(b) Find the lowest common multiple (LCM) of $2A$ and $3B$.
Give the LCM as a product of powers of its prime factors.

.....

(Total for Question 9 is 4 marks)

Q32**PAST PAPER**

Edexcel 4MA1 • January 2021 • Paper 2H • Q9

CH 1 • TOPIC 3 2 marks

9

$$A = 2^3 \times 3^2 \times 5^2 \times 11$$

$$B = 2^4 \times 3 \times 5^4 \times 13$$

Find the lowest common multiple (LCM) of A and B .

Give your answer as a product of powers of prime numbers.

(Total for Question 9 is 2 marks)**Q33****PAST PAPER**

Edexcel 4MA1 • January 2022 • Paper 1HR • Q6

CH 1 • TOPIC 3 4 marks

6 (a) Work out the lowest common multiple (LCM) of 36 and 120

$$A = 5^2 \times 7^4 \times 11^p$$

$$B = 5^m \times 7^{n-5} \times 11$$

 m , n and p are integers such that

$$m > 2$$

$$n > 10$$

$$p > 1$$

(b) Find the highest common factor (HCF) of A and B

Give your answer as a product of powers of its prime factors.

(Total for Question 6 is 4 marks)

Q34 PAST PAPER Edexcel 4MA1 • June 2022 • Paper 1HR • Q12

CH 1 • TOPIC 3 3 marks

$$12 \quad P = 3^3 \times 5^2 \times 7$$

$$Q = 3^2 \times 5 \times 7^2$$

(a) Write down the highest common factor (HCF) of P and Q

$$P = 3^3 \times 5^2 \times 7$$

$$Q = 3^2 \times 5 \times 7^2$$

(b) Work out the value of $P^3 \times Q$ Give your answer in the form $3^x \times 5^y \times 7^z$ where x , y and z are positive integers.

(Total for Question 12 is 3 marks)

Q35 PAST PAPER Edexcel 4MA1 • November 2023 • Paper 1H • Q7

CH 1 • TOPIC 3 2 marks

$$7 \quad A = 5^3 \times 7^3 \times 11^6 \quad \text{and} \quad B = 5^6 \times 7^2 \times 11^4$$

Find the highest common factor (HCF) of A and B

Give your answer as a product of powers of its prime factors.

(Total for Question 7 is 2 marks)

Q36**PAST PAPER**

Edexcel 4MA1 • May 2024 • Paper 1HR • Q11

CH 1 • TOPIC 3 4 marks

11 $A = 2^5 \times 5 \times 7^2$

$B = 2^3 \times 5^3 \times 7^4$

- (a) Write down the highest common factor (HCF) of
- $5A$
- and
- $2B$

Give your answer as a product of prime factors.

.....
(2)

$A = 2^5 \times 5 \times 7^2$

$B = 2^3 \times 5^3 \times 7^4$

- (b) Work out the value of
- $(AB)^2$

Give your answer as a product of prime factors.

.....
(Total for Question 11 is 4 marks)

Q37**PAST PAPER**

Edexcel 4MA1 • November 2024 • Paper 1H • Q6

CH 1 • TOPIC 3 2 marks

6 $A = 2^3 \times 5^4 \times 7 \times 11$

$$B = 2^2 \times 5^2 \times 7^2$$

$$C = 2^2 \times 5^3 \times 7^4$$

Find the highest common factor (HCF) of A , B and C
Write your answer as a product of prime factors.

.....
(Total for Question 6 is 2 marks)

04 LCM Problems: “Together Again”

Two things repeat at different intervals and start together — they meet again after the LCM. Buy “the same number of each” — the LCM again.

WHEN YOU SEE → YOU DO

“every 24 minutes ... every 20 minutes ... next leave together”

→ $\text{LCM}(24, 20) = 120$ min, add to the start time

“exactly the same number of pens as pencils”

→ LCM of the pack sizes, then count the packs

“more than 60 of each”

→ first multiple of the LCM above 60

“how many times in one hour”

→ $60 \text{ min} \div \text{LCM}$

“state an assumption”

→ they keep to the timetable exactly

Find the LCM

Answer the real question

WORKED MODEL • JUNE 2023 P2HR

Pens: packets of 9 at £7.60. Pencils: boxes of 12 at £4.80. Same number of pens as pencils. Minimum cost?

1 $\text{LCM}(9, 12) = 36$

same number of each

2 $36 \div 9 = 4$ packets, $36 \div 12 = 3$ boxes

count the packs

3 $4 \times 7.60 + 3 \times 4.80 = 30.40 + 14.40$

cost

ANSWER £44.80

IDEA 04

Practise: LCM Problems: Together Again

My questions first, then selected practice, then real past papers.

Q38 **DR ESLAM** Together againCH 1 • TOPIC 3 **3 marks**

Two lights flash every 15 seconds and every 20 seconds. They flash together at 9:00:00. When do they next flash together?

SELECTED PRACTICE**Q39** **PRACTICE** Practice • EasyCH 1 • TOPIC 3 **3 marks**

4 Trams leave Piccadilly

to Eccles every 9 minutes
to Didsbury every 12 minutes

A tram to Eccles and a tram to Didsbury both leave Piccadilly at 9 am.
At what time will a tram to Eccles and a tram to Didsbury next leave Piccadilly at the same time?

(3 marks)

Q40**PRACTICE**

Practice • Medium

CH 1 • TOPIC 3**3 marks**

3 Matt and Dan cycle around a cycle track.

Each lap Matt cycles takes him 50 seconds.

Each lap Dan cycles takes him 80 seconds.

Dan and Matt start cycling at the same time at the start line.

Work out how many laps they will each have cycled when they are next at the start line together.

(3 marks)

Q41**PRACTICE**

Practice • Medium

CH 1 • TOPIC 3 5 marks

16 (a) A bus timetable shows the following information.

- A bus following route T leaves for the train station every 20 minutes.
- A bus following route A leaves for the airport every 18 minutes.
- A bus following route T and a bus following route A both leave at 8.37 am.

When is the next time one of each bus is timetabled to leave at the same time?

(4 marks)

(b) Write down one assumption that was necessary to solve this problem.

(1 mark)

Q42**PRACTICE**

Practice • Hard

CH 1 • TOPIC 3 3 marks

5 Liz buys packets of coloured buttons.

There are 8 red buttons in each packet of red buttons.

There are 6 silver buttons in each packet of silver buttons.

There are 5 gold buttons in each packet of gold buttons.

Liz buys equal numbers of red buttons, silver buttons and gold buttons.

How many packets of each colour of buttons did Liz buy?

..... packets of red buttons

..... packets of silver buttons

..... packets of gold buttons

(3 marks)

PAST PAPERS

Q43**PAST PAPER**

Edexcel 4MA1 • June 2023 • Paper 2HR • Q2

CH 1 • TOPIC 3**4 marks**

- 2 Sandeep wants to buy some packets of pens and some boxes of pencils for his stationery shop.

Each packet of pens contains 9 pens.

Each box of pencils contains 12 pencils.

Each packet of pens costs £7.60

Each box of pencils costs £4.80

Sandeep can only buy full packets of pens and full boxes of pencils.

He wants to buy exactly the same number of pens as pencils.

Work out the minimum amount Sandeep needs to pay.

£.....

(Total for Question 2 is 4 marks)

ALL The Whole Topic on One Page

Tear this page out before the exam.

ZONE HCF

shared primes

lowest powers — “share into equal groups”

ZONE LCM

every prime

highest powers — “together again”,
“same number of each”

- | | | |
|-----------|----------------------|--|
| 01 | Prime factors | Tree until every end is prime; write as powers in order. |
| 02 | Venn method | Middle = HCF; everything = LCM. |
| 03 | Index form | HCF lowest shared powers; LCM highest powers of all primes. |
| 04 | Problems | “Together again” / “same number of each” = LCM, then answer the story. |

BOTTOM LINE Primes first → HCF = shared, lowest → LCM = all, highest.

Answers

Prime Factors, HCF and LCM · for the teacher. Every answer checked with a computer algebra system.

IDEA 01 • PRODUCT OF PRIME FACTORS

Q1	Dr Eslam · Factor tree	$2 \times 3^2 \times 5$
Q2	Dr Eslam · Index form	$2^3 \times 3^2 \times 5$
Q3	Practice · Easy	$3 \times 5^2 \times 7$
Q4	Practice · Easy	$2^{n+3} \times 3 \times 5^m$
Q5	Practice · Easy	$2^3 \times 3^2 \times 7$
Q6	Practice · Hard	$2^2 \times 3^9 \times y^6$
Q7	Edexcel 4MA1 · January 2020 · Paper 2HR · Q1	(a) 5^{19} (b) $2^5 \times 5^2$
Q8	Edexcel 4MA1 · May 2021 · Paper 2H · Q1	$2^3 \times 3 \times 5^2$
Q9	Edexcel 4MA1 · November 2021 · Paper 1H · Q7	$2^4 \times 3^2 \times 5^2$
Q10	Edexcel 4MA1 · June 2022 · Paper 2HR · Q2	$2^4 \times 3 \times 5^2$
Q11	Edexcel 4MA1 · May 2023 · Paper 1HR · Q1	$2 \times 3^2 \times 5^3$
Q12	Edexcel 4MA1 · June 2024 · Paper 2HR · Q1	$2^3 \times 5^2 \times 7$
Q13	Edexcel 4MA1 · June 2025 · Paper 2HR · Q2	$2^3 \times 5^2 \times 7$

IDEA 02 • HCF AND LCM OF NUMBERS

- Q14** Dr Eslam • Venn method (a) 12 (b) 720
- Q15** Practice • Easy 35
- Q16** Practice • Medium 18
- Q17** Practice • Medium (a) $2^2 \times 3^2 \times 7$ (b) 84
- Q18** Practice • Hard 36 and 72
- Q19** Edexcel 4MA1 • January 2020 • Paper 1H • Q2 420
- Q20** Edexcel 4MA1 • January 2022 • Paper 2H • Q7 252
- Q21** Edexcel 4MA1 • June 2022 • Paper 1H • Q3 (a) 28 (b) 360
- Q22** Edexcel 4MA1 • May 2023 • Paper 1H • Q8 (a) $2^2 \times 3 \times 5^2$ (b) $2^3 \times 3^3 \times 5^2 \times 7 = 37\,800$
- Q23** Edexcel 4MA1 • May 2024 • Paper 1H • Q3 36
- Q24** Edexcel 4MA1 • June 2025 • Paper 1HR • Q1 630

IDEA 03 • HCF AND LCM IN INDEX FORM

- Q25** Dr Eslam • Lowest, highest HCF $2^2 \times 3$; LCM $2^4 \times 3^3 \times 5^2 \times 7$
- Q26** Practice • Medium (a) $3^2 \times 5^3 \times 7$ (b) $3^4 \times 5^4 \times 7 \times 11$
- Q27** Practice • Medium (a)(i) $2^3 \times 3 \times 5$ (ii) 9 (b) $2^3 \times 3 \times 7 = 168$
- Q28** Practice • Hard $2 \times 3^{11} \times 5 = 1\,771\,470$
- Q29** Practice • Hard 98^{91}
- Q30** Edexcel 4MA1 • Oct/Nov 2020 • Paper 1H • Q10 (a) 2×3^{37} (b) $2^5 \times 3^{80}$
- Q31** Edexcel 4MA1 • January 2021 • Paper 1HR • Q9 (a) $2^6 \times 3 \times 11^4$ (b) $2^9 \times 3^5 \times 11^8$
- Q32** Edexcel 4MA1 • January 2021 • Paper 2H • Q9 $2^4 \times 3^2 \times 5^4 \times 11 \times 13$
- Q33** Edexcel 4MA1 • January 2022 • Paper 1HR • Q6 (a) 360 (b) $5^2 \times 7^4 \times 11$
- Q34** Edexcel 4MA1 • June 2022 • Paper 1HR • Q12 (a) $3^2 \times 5 \times 7$ (b) $3^{11} \times 5^7 \times 7^5$
- Q35** Edexcel 4MA1 • November 2023 • Paper 1H • Q7 $5^3 \times 7^2 \times 11^4$
- Q36** Edexcel 4MA1 • May 2024 • Paper 1HR • Q11 (a) $2^4 \times 5^2 \times 7^2$ (b) $2^{16} \times 5^8 \times 7^{12}$
- Q37** Edexcel 4MA1 • November 2024 • Paper 1H • Q6 $2^2 \times 5^2 \times 7$

IDEA 04 • LCM PROBLEMS: TOGETHER AGAIN

- Q38** Dr Eslam • Together again LCM = 60 s: 9:01:00
- Q39** Practice • Easy LCM 36: 9:36 am
- Q40** Practice • Medium LCM 400 s: Matt 8 laps, Dan 5 laps
- Q41** Practice • Medium (a) LCM 180 min: 11:37 am (b) the buses leave exactly on time
- Q42** Practice • Hard 15 red, 20 silver, 24 gold
- Q43** Edexcel 4MA1 • June 2023 • Paper 2HR • Q2 £44.80